

Element H

The design we chose consists of 2 PVC pipes connecting two wooden handles that hold a PVC handlebar. The design uses hot glue to hold the PVC and wood in place along with the $\frac{1}{2}$ " elbow fittings for the 90 degree angles. The mechanism utilizes as little material as possible for the functionality and lightweight functions.

Two high-priority requirements that we focused on for this design were the removability factor and the adjustability. We felt that having these two concepts incorporated in our train of thought would provide us with the best design under the budget constraints. The adjustability constraints were vast as we had to create a design ideal for a 6'3" client, but one that was also usable at other heights including 6' and below as well.

There were a series of tests that took place to test these criteria. Obviously, tests were run each day to test consistency and prioritize the focus on continuing to improve the design. The removability was one of our biggest focuses because we felt this priority was one of the more important criteria. The ability to quickly remove the device and move it to another cart was important for us. Therefore, our design revolved around a lightweight design that needed little to remove, but would have enough friction when placed to stay in place. We ran tests similar to Joseph Johnson, in that we tested how long it would take to attract and remove the device from the cart. Using the suggestions from the Vanderbilt professors and Mr. Kiser we made specific modifications to our design to make our design better.

Evaluating the design we improved the comfort ratings, and the durability by adding the hot glue and PVC fittings. Overall the modifications included adding handles supported by cloth and other materials comfortable for the hands.

The product works under extreme circumstances; however, certain happenings may put too much pressure on the design causing it to bend or break. To further improve the design we would test the design under more severe conditions to test whether the design would withstand the pressure or fall to the stress of the weight.

Final Design:



Secondary Prototype:

